Properties of vector (cross) product.

(i) The vector product of any two vectors is always another vector whose direction is perpendicular to the plane containing these two vectors, i.e., orthogonal to both the vectors $\vec{A}$ and $\vec{B}$, even though the vectors $\vec{A}$ and $\vec{B}$ may or may not be mutually orthogonal.

(ii) The vector product of two vectors is not commutative, i.e., $\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$ But., $\vec{A}\times\vec{B} = -[\vec{B}\times\vec{A}]$
Here it is worthwhile to note that $|\vec{A}\times\vec{B}| = |\vec{B}\times\vec{A}| = AB\sin\theta$ i.e., in the case of the product vectors $\vec{A}\times\vec{B}$ and $\vec{B}\times\vec{A}$, the magnitudes are equal but directions are opposite to each other.

(iii) The vector product of two vectors will have maximum magnitude when $\sin\theta = 1$, i.e., $\theta = 90°$
i.e., when the vectors $\vec{A}$ and $\vec{B}$ are orthogonal to each other.

$$\left(\vec{A}\times\vec{B}\right)_{max} = AB\hat{n}$$

(iv) The vector product of two non–zero vectors will be minimum when $|\sin\theta| = 0$, *i.e.*, $\theta = 0°$ *or* $180°$

$$\left(\vec{A}\times\vec{B}\right)_{min} = 0$$

i.e., the vector product of two non–zero vectors vanishes, if the vectors are either parallel or antiparallel.

(v) The self–cross product, i.e., product of a vector with itself is the null vector

$$\vec{A}\times\vec{A} = AA\sin 0°\hat{n} = \vec{0}.$$

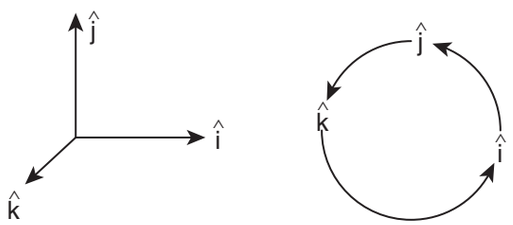
In physics the null vector $\vec{0}$ is simply denoted as zero.

(vi) The self–vector products of unit vectors are thus zero.

$$\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \vec{0}$$

(vii) In the case of orthogonal unit vectors, $\hat{i},\hat{j},\hat{k}$, in accordance with the right hand screw rule:

$$\hat{i}\times\hat{j} = \hat{k},\ \hat{j}\times\hat{k} = \hat{i}\ \textit{and}\ \hat{k}\times\hat{i} = \hat{j}$$



Also, since the cross product is not commutative,

$$\hat{j}\times\hat{i} = -\hat{k}, \hat{k}\times\hat{j} = -\hat{i}$$
$$\textit{and}\ \hat{i}\times\hat{k} = -\hat{j}$$

(viii) In terms of components, the vector product of two vectors $\vec{A}$ *and* $\vec{B}$ is

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

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